

Orbital Period

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9:27 AM

Specific energy of any body:

$$E_s = \frac{v^2}{2} - \frac{Gm_{\text{sun}}}{r}$$

The semi-major axis is determined by energy:

$$a = - \frac{Gm_{\text{sun}}}{2E_s}$$

The total orbital period then

becomes

$$T = 2\pi \sqrt{\frac{a^3}{Gm_{\text{sun}}}}$$